V.I.P.E.R.

VIRTUAL INTELLIGENCE PLATFORM FOR EVIDENCE & RECORDS

Supervisor-Link

Network & Firewall Configuration Guide

Prepared for IT / Network Administrators

WHAT THIS ENABLES

Lets the Officer application discover and securely connect to the V.I.P.E.R. Supervisor dashboard running on another PC on the LAN.

PORTS TO ALLOW (on the Supervisor PC)

TCP 7071	Secure WebSocket link
UDP 7071	LAN auto-discovery (broadcast)

Hand this document to your IT department.

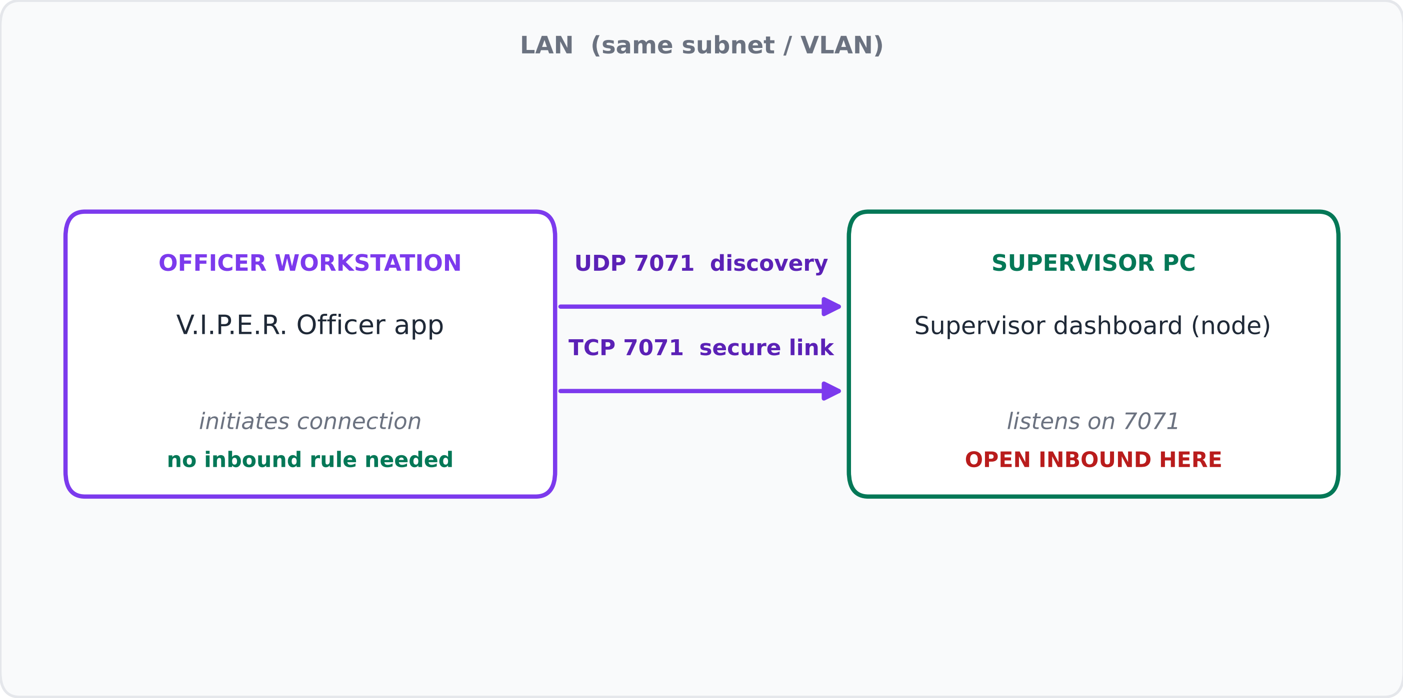
No changes are required on the officer's workstation for a basic connection — the inbound rules below are applied on the Supervisor host.

Document version 1.0 · Applies to V.I.P.E.R. app v4.x · Default port 7071

How the connection works

The Officer app (investigator workstation) talks to the Supervisor dashboard over your local network using two channels on the same port number, 7071:

- Discovery (UDP 7071): the Officer app sends a broadcast asking "are there any Supervisor nodes here?" Each Supervisor replies with its address. This is how the Scan button auto-populates nodes.
- Secure link (TCP 7071): once a node is chosen, an encrypted WebSocket connection (ws://<supervisor-ip>:7071) carries the actual handshake and data.



KEY POINT

Only the Supervisor PC needs inbound firewall rules. The officer workstation makes an outbound connection, which Windows normally allows by default. UDP broadcast discovery does not cross subnets/VLANs — if the two PCs are on different networks, discovery finds 0 nodes and you must enter the address manually (see p.6).

REQUIREMENTS

What to open, and where

Apply the following on the machine running the Supervisor dashboard. These are the only rules required for officers to discover and connect.

PORT	PROTOCOL	DIRECTION	PURPOSE
7071	TCP	Inbound	Secure WebSocket link (data + handshake)
7071	UDP	Inbound	LAN auto-discovery reply (broadcast)

Scope of the rules

- Profile: apply to the Private (Domain) profile that matches your agency LAN. Avoid the Public profile unless required.
- Program: you may scope the rules to the V.I.P.E.R. Supervisor executable for tighter security, or leave port-based if the node runs under Node/Electron.
- Remote scope (optional hardening): restrict the allowed remote addresses to your officer subnet (e.g. 10.0.0.0/24) instead of Any.

Changing the default port (optional)

7071 is the default. If it conflicts with another service, the Supervisor host can override it with environment variables before launch — then open the new port instead:

```
set VIPER_SUPERVISOR_DISCOVERY_PORT=7071
set VIPER_SUPERVISOR_LAN_URL=ws://0.0.0.0:7071
```

Add the inbound rules

OPTION A — GUI Windows Defender Firewall with Advanced Security

1. Press Win+R, type wf.msc and press Enter.
2. Select "Inbound Rules" in the left pane, then "New Rule…" on the right.
3. Rule Type: choose "Port" → Next.
4. Protocol/Ports: select "TCP", enter Specific local port 7071 → Next.
5. Action: "Allow the connection" → Next.
6. Profile: tick the profile(s) for your LAN (usually Domain and/or Private) → Next.
7. Name it "VIPER Supervisor-Link (TCP 7071)" → Finish.
8. Repeat steps 2–7, but at Protocol choose "UDP" (name it "… (UDP 7071)").

OPTION B — COMMAND LINE Run an elevated (Administrator) Command Prompt

```
netsh advfirewall firewall add rule name="VIPER Supervisor-Link TCP 7071" ^
   dir=in action=allow protocol=TCP localport=7071 profile=domain,private

netsh advfirewall firewall add rule name="VIPER Supervisor-Link UDP 7071" ^
   dir=in action=allow protocol=UDP localport=7071 profile=domain,private
```

TIP PowerShell equivalent: New-NetFirewallRule -DisplayName 'VIPER Supervisor-Link TCP 7071' -Direction Inbound -Protocol TCP -LocalPort 7071 -Action Allow (repeat with -Protocol UDP).

Third-party AV, network & hardware firewalls

Many agency workstations run additional security layers that can block port 7071 even after Windows Defender is configured. Check each that applies:

Endpoint security / antivirus suites

- Products such as CrowdStrike, SentinelOne, Symantec, McAfee, Sophos, ESET and Windows Defender for Endpoint have their own firewall/network-control modules that override the OS firewall.
- In the product console, add an inbound allow rule (or application exception) for TCP+UDP 7071 on the Supervisor host.
- Watch for "network intrusion prevention" / "exploit protection" modules that silently drop LAN broadcasts — whitelist the app.

Network firewalls, switches & VLANs

- Perimeter/segmentation firewalls (Palo Alto, Fortinet, Cisco ASA, SonicWall) must permit TCP 7071 between the officer and supervisor subnets.
- UDP broadcast discovery does NOT cross routers/VLANs. If officers and supervisors are on different VLANs, expect discovery to return 0 nodes — use manual address entry, or ask networking to enable a directed-broadcast/relay if auto-discovery is required.
- Wireless "client isolation" / "AP isolation" on guest or staff Wi-Fi blocks device-to-device traffic; disable it for the relevant SSID/VLAN.

Group Policy (managed fleets)

- If workstations are domain-managed, local firewall changes may be reverted by GPO. Have the domain admin publish the two inbound rules via Group Policy so they persist.
- Confirm "Windows Firewall: Protect all network connections" policy is not forcing settings that strip the rule.

VERIFY & TROUBLESHOOT

Reading the built-in diagnostics

In the Officer app open Settings → Supervisor Link → "Run connection diagnostics". The report's VERDICT tells you exactly where it broke. Decoder:

ENOTFOUND

What it means: the address is malformed / not resolvable — a bad hostname, not a firewall.

Fix: use the exact form ws://<ip>:7071 with a COLON before the port (e.g. ws://192.168.1.50:7071, not ws://192.168.1.50.7071).

ECONNREFUSED

What it means: address is valid and reachable, but nothing is listening / the port is closed.

Fix: confirm the Supervisor dashboard is running, and that inbound TCP 7071 is allowed on that PC.

TIMEOUT / CONNECT_TIMEOUT

What it means: packets are being silently dropped — the classic firewall symptom.

Fix: open TCP 7071 in Windows Defender AND any AV / network firewall between the two machines (pp. 4–5).

discovery 0 node(s)

What it means: no Supervisor replied to the UDP broadcast (blocked, or different subnet/VLAN).

Fix: allow inbound UDP 7071 on the Supervisor PC; if on different subnets, enter the ws:// address manually.

Secure link OK

What it means: TCP + WebSocket + handshake all passed.

Fix: you're connected. If identity shows NOT SET, pair/enroll the officer identity in Supervisor settings.

QUICK MANUAL TEST (from the officer PC)

```
Test-NetConnection -ComputerName <supervisor-ip> -Port 7071
```